Pierre Guilbert

Computer Vision, Machine Learning & Mathematical Modelling
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Engineer with 10 years of experience in Computer Vision, Machine Learning, and Mathematical Modelling, with a strong interest in differential geometry and nonlinear optimization.

Passionate about applying advanced mathematics to real-world challenges, I enjoy combining technical expertise and creative problem-solving through collaboration and intellectual exchange.

Experience

Genius Sports — Senior Computer Vision Engineer 2019 - Present

Lausanne

- Camera modelling and calibration: keypoint detection, bundle adjustment, and optical SLAM.
- Human pose estimation: body keypoint detection, triangulation, and trajectory optimization based on physical laws and optical measurements.
- Geometric and physical parameter estimation from multi-view images: 3D reconstruction, Gaussian splatting, ball spin, and ball deformation.
- Ad-tech: auction price optimization and user click-through prediction.

Kitware — Computer Vision Engineer 2016 - 2019

Lyon

- Software engineering for VeloView: lidar, multi-sensor fusion, and point-cloud processing.
- Lidar-based SLAM: design and implementation of a lidar-based SLAM system inspired by the LOAM algorithm.
- Contributions to open-source projects: VTK, ParaView, and Ceres Solver.

CNRS — Image Processing Research Assistant 2015 - 2016

Saint-Étienne

- Research in image and signal processing using the logarithmic image processing (LIP) model.
- Neural network training framework using LIP operations.

Skills

Mathematical Modelling: Physics, PDEs, variational calculus, nonlinear optimization

Computer Vision: Multi-view geometry, bundle adjustment, SLAM, pose estimation

Machine Learning: Design and training of discriminative and generative models (classification, segmentation, regression, density estimation) with PyTorch

Mathematics: Differential geometry, topology, measure theory, algebra

Programming: C++ (Eigen, Ceres, OpenCV, PCL), Python, Git

Education

Graduate studies in Mathematics & Engineering — Mines Saint-Étienne 2016

MSc Computer Graphics & Signal Processing — Université Lyon 1 2015

MEng Electrical Engineering — Université Lyon 1 2015

BSc Mathematics, Physics & Chemistry — CPGE Champollion 2012

Personal Interests

Philosophy & Literature: Exploring morality, ethics, consciousness, and the nature of reality

Stock market investment: Long-term stock picking with no intention to outperform the market, but to learn from it

Powerlifting: Strength training in bench press, deadlift, and squat

Mathematics: Exploring the state of the art, writing research notes and short mathematical treatises